An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss forces internal backpropagation gradients to align directly with true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

The digital transformation of tertiary healthcare facilities has produced cyber-physical ecosystems where uninterrupted electrical supply is not merely an op-

erational convenience but a patient-safety imperative. Modern smart hospitals operate localized microgrids that integrate utility feeders, rooftop photovoltaic arrays, battery energy storage, and emergency diesel generators to power life-critical loads: mechanical ventilators and extracorporeal membrane oxygenation circuits in intensive-care units, robotic surgical suites requiring sub-second voltage stability, mRNA vaccine cold-chains maintained at $-80°$C, and Internet-of-Medical-Things (IoMT) edge gateways linking bedside monitors to electronic health record servers [5,19]. An undetected voltage collapse or spurious load-shedding command initiated by a cyberattack on this infrastructure does not merely incur financial loss—it threatens lives.

Supervisory Control and Data Acquisition (SCADA) systems stream bus-level measurements—voltage magnitude V, phase angle θ, active power P, reactive power Q—from Remote Terminal Units (RTUs) into Weighted Least Squares (WLS) state estimators several times per minute. The classical bad-data detector applies a χ^2 residual threshold to flag corrupted telemetry. Liu et al. [15] demonstrated that an adversary with partial knowledge of the network admittance matrix can construct a stealthy attack vector $a = Hc$, where H is the measurement Jacobian and c is the targeted state bias, such that the WLS residual remains unchanged. The injected corruption cancels inside the χ^2 test, leaving the state estimator converging to a biased solution while hospital operators remain unaware [13].

Early data-driven methods for FDIA detection relied primarily on multi-layer perceptrons (MLPs) and convolutional networks [12,3]. While effective under fixed network conditions, these models assume static measurement vectors. In hospital distribution systems, line switching, breaker trips, and maintenance reconfigurations alter active power paths. Under these events, the spatial mapping between physical sensors and input indices changes, leading static classifiers to trigger false alarms. Re-training models after each switching event is computationally costly and impractical for real-time substation operation.

To address variable network topologies, graph neural networks (GNNs) represent power networks directly, treating substations as nodes and power lines as edges [21]. In this formulation, disconnected lines correspond to edge removals in the graph adjacency, preserving node feature dimensions. Boyaci et al. [2] implemented ARMA graph wavelets for joint detection and localization, while Takiddin et al. [22] confirmed GNN generalization across diverse feeder topologies. However, conventional graph convolutions apply isotropic aggregation across neighboring buses, which can dilute localized measurement spikes when substations are closely coupled electrically.

In addition, practical deployment in healthcare facilities requires meeting two operational constraints. First, detection models must provide interpretable outputs within SCADA polling periods (typically 2–4 seconds). Perturbation-based tools such as SHAP or LIME require numerous model evaluations and are too slow for real-time operation [27]. Second, detecting an attack without automated remediation leaves patient-critical loads exposed. A closed-loop mechanism is required to isolate compromised channels and reconstruct the true electrical states

before automated control systems issue erroneous dispatch or load-shedding commands.

1.1 Contributions

To address these challenges, this paper presents an explainable cognitive graph digital twin (CGDT) for smart hospital microgrids. The primary contributions are:

- A state-space synchronization formulation that tracks SCADA measurements using steady-state Kalman gain K_{sync} alongside an exponential rolling baseline to maintain nominal electrical references.
- An admittance-conditioned GraphSAGE layer incorporating complex line parameters $Y_{uv} = G_{uv} + jB_{uv}$ directly into the aggregation function to preserve circuit physics.
- An attribution-guided multi-task loss ($\mathcal{L}_{\text{align}}$) that uses Integrated Gradients during training to suppress gradient leakage into benign buses, improving localization F1 by 5.52 pp.
- An automated mitigation protocol that isolates compromised RTUs and reconstructs telemetry via admittance-weighted interpolation, reducing post-attack WLS error by 99.97% in under 2 ms.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 formulates the AC threat model; Section 4 describes the CGDT architecture; Section 5 details feature attribution and automated mitigation; Sections 6 and 7 present experimental benchmarks on the IEEE 14-bus and 30-bus systems; and Section 8 concludes the paper.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

The integration of clinical IT networks with operational technology (OT) in modern hospitals expands exposure to cyber-physical threats [11,5]. In particular, hospital microgrids feed loads with strict continuity requirements, including surgical suites and intensive care units. In their survey of medical cyber-physical systems, Sun et al. [19] identified SCADA telemetry manipulation as a severe hazard. While prior studies have addressed healthcare network security using Bayesian risk assessments [4] or IoT monitoring frameworks [17], most existing methods process telemetry as independent time-series signals without modeling the underlying power distribution topology.

2.2 False Data Injection Attacks in Cyber-Physical Energy Systems

False data injection attacks against power grid state estimation were introduced by Liu et al. [15], who showed that an attacker with knowledge of the measurement Jacobian H can craft attack vectors $a = Hc$ that leave WLS residuals

unchanged. Subsequent research established that stealthy attacks are feasible in AC systems, under partial network observability, and in distributed architectures [13,6]. Analogous vulnerabilities have also been examined in DC microgrids [3,18]. To detect stealthy injections, machine learning techniques have been investigated, including support vector classifiers, deep MLPs [12], and recurrent neural networks [14]. However, these approaches rely on fixed input representations, making them sensitive to routine topology reconfigurations.

2.3 Graph Neural Networks for Grid Security

GNNs represent power networks by treating substations as nodes and transmission branches as edges [21]. Early applications employed spectral graph convolutions [10] or attention mechanisms [25]. To scale to larger systems, Hamilton et al. [8] developed GraphSAGE, which aggregates local neighborhood samples and concatenates center-node representations. In power grid cyber-defense, Boyaci et al. [2] used ARMA graph filters to achieve 96.25% detection F1, and Takiddin et al. [22] and Yue and Qu [28] confirmed the flexibility of GNNs across variable network structures. Luo et al. [16] investigated federated GNN implementations for telemetry privacy. Nevertheless, most existing architectures use unweighted or topological adjacency without incorporating physical line admittance into message passing, limiting their ability to capture localized electrical disturbances.

2.4 Explainability and Digital Twins in Critical Infrastructure

For mission-critical applications, security models should provide interpretable explanations for generated alarms. Perturbation-based explainability tools like LIME or SHAP require hundreds of model evaluations, making them computationally impractical for online SCADA cycles [27,26]. In contrast, Integrated Gradients (IG) [20] computes feature attributions via path integrals in a single backward pass and satisfies the completeness axiom. Meanwhile, digital twin frameworks [7,23] have been proposed for power system monitoring [1], and Kamel Boulos and Zhang [9] discussed digital twins for healthcare environments. In this paper, we combine state-space synchronization with IG-based attribution to provide both real-time monitoring and verifiable anomaly localization at the local substation level.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N \times N}$ defines nodal

conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big) \tag{1}$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1} [z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H \big)^{-1} H^T R^{-1} \big[z - h(x^{(k)}) \big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x) / \partial x$ is the measurement Jacobian. Conventional bad-data detection flags potential measurement anomalies when the normalized residual energy $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ exceeds a threshold τ_{χ^2}, determined from a chi-square distribution with degrees of freedom equal to the number of measurements minus system states.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. Using knowledge of the network admittance structure, the attacker constructs the injection vector according to:

$$a = Hc \tag{4}$$

When injected into the communication stream, the received telemetry becomes $z_{\text{comp}} = z + a = z + Hc$. The resulting state estimate shifts to $\hat{x}_{\text{comp}} = \hat{x} + c$. Evaluating the post-attack residual yields:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &= \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \leq \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the perturbation vector cancels algebraically within the residual calculation ($r_{\text{comp}} = r$), the attack leaves the χ^2 detector unaffected ($\tau_{\chi^2} = 23.68$ for df $= 14$ at $\alpha = 0.05$). Consequently, classical state estimation accepts the corrupted values without triggering an alert.

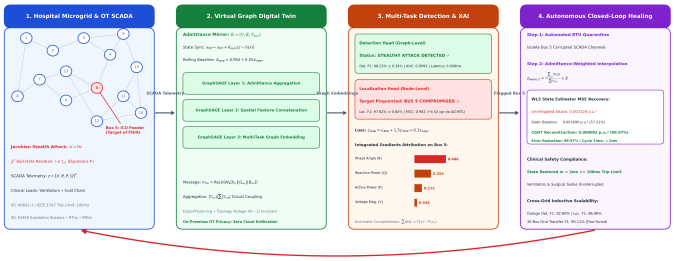


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

Clinical Impact in Hospital Microgrids. In healthcare facilities, distribution substations supply power to departments with critical life-support loads, such as operating rooms and intensive care wards. Unidentified state bias on these feeders can mislead energy management systems into executing inappropriate control actions, such as incorrect transformer tap changes or unnecessary load curtailment. Such events risk interrupting ventilator operation or cooling in vaccine storage facilities, introducing patient-safety risks that standard economic loss models do not address.

4 The Cognitive Graph Digital Twin Architecture

Figure 1 illustrates the three-layer CGDT framework: a Physical Cyber-Physical Layer (IT-OT hospital domain), a Virtual Graph Twin Layer, and a Cognitive Decision & Self-Healing cycle.

4.1 Digital Twin State-Space Synchronization

SCADA telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ arrives at sampling period T_s via IEC 61850 / 5G. The virtual twin updates its mirror state via tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P is state-prediction covariance. A dynamic rolling baseline maintains the nominal operating envelope:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

using verified clean telemetry z_{clean}. Removing this baseline drops localization F1 by 16.32 pp (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each measurement interval, telemetry from all monitored substations is structured into an attributed graph $G = (V, E, X, E_{\text{attr}})$:

- *Node Features*: Each bus $i \in V$ receives state vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$, yielding matrix $X \in \mathbb{R}^{N\times 4}$ normalized across historical operating ranges.
- *Edge Attributes*: For each branch $(i, j) \in E$, attribute vector $[G_{ij}, B_{ij}]^T$ stores line conductance and susceptance from admittance matrix Y_{bus}, forming edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$.

When a transmission branch trips, the corresponding edge is removed from E. Because node dimensions remain fixed while the edge set updates, the model handles topology adjustments without requiring retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE [8] averages neighbor embeddings across topological connections without accounting for branch electrical characteristics. To incorporate physical coupling, the message function is conditioned on complex line admittance $Y_{uv} = G_{uv} + jB_{uv}$:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes vector concatenation, and W_e and W_k are trainable weight matrices. Normalizing message weights by $|Y_{uv}|/\sum_w |Y_{vw}|$ scales neighbor influence according to electrical admittance, reflecting Ohm's and Kirchhoff's laws in the message passing. Concatenating the previous center-node embedding $h_v^{(k-1)}$ retains bus-specific features, mitigating the over-smoothing common to isotropic GNNs.

4.4 Multi-Task Detection and Localization Heads

The shared backbone embedding H feeds two prediction heads:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \text{Softmax}(W_{\text{node}}\, H) \tag{10}$$

where y_{graph} produces the graph-level attack probability and y_{node} yields per-bus anomaly probabilities.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

In conventional multi-task GNNs, gradient attributions are computed solely after training for interpretability. Here, we integrate feature attribution directly into the training loss to improve bus localization. For each attacked training sample, Integrated Gradients computes an attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$. We define an attribution alignment loss:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ is the ℓ_2-norm of the attribution vector at bus v, reflecting model sensitivity to that bus, and $\hat{y}_{\text{node},v} \in \{0, 1\}$ is the ground-truth node label. When the network assigns high attribution to unattacked buses, $\mathcal{L}_{\text{align}}$ penalizes the configuration. The composite training loss is:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy, $\mathcal{L}_{\text{WCE}}$ is weighted cross-entropy to handle class imbalance (as attacked buses are sparse), and $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ are balancing hyperparameters selected empirically.

4.6 On-Premises Substation Deployment

Hospital electrical telemetry contains facility-specific operational data, including wing-level load variations and medical equipment usage schedules. Transmitting raw measurements to external cloud servers introduces privacy and regulatory compliance challenges under health data frameworks. The CGDT architecture is designed for local edge deployment within the hospital substation OT network. Telemetry processing, threat classification, and state reconstruction are executed on-premises, avoiding the need to export raw measurement streams to third-party infrastructure.

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

To assist grid operators in diagnosing detected events, the CGDT computes feature attributions using Integrated Gradients (IG) [20]. Given a nominal reference state x_0 and an incoming measurement vector x, the attribution of feature i is calculated by integrating gradients along the straight-line interpolation path:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated with $S = 50$ Riemann increments. IG satisfies the completeness axiom ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and implementation invariance. For the IEEE 14-bus system, this calculation executes in 0.81 ms, allowing near real-time attribution within SCADA polling cycles. In our case study of a stealthy attack at Bus 5 (Section 7.1), the attribution module identified voltage phase angle ($\theta = 0.486$) as the primary contributing feature.

5.2 Automated Sensor Quarantine and Spatial Reconstruction

When the node localization head identifies bus k as compromised ($y_{\text{node},k} > \gamma_{\text{loc}} = 0.5$), the system initiates an automated remediation sequence:

1. *Channel Exclusion*: Measurements from bus k (V_k, θ_k, P_k, Q_k) are marked untrusted and excluded from the active WLS measurement buffer.
2. *Spatial Interpolation*: Rather than substituting static historical averages, excluded telemetry is reconstructed by interpolating measurements from healthy adjacent buses, weighted by line admittance:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta \sim \mathcal{U}(\pm 2\%) \tag{14}$$

   where $|Y_{kj}|$ represents admittance magnitude and δ models measurement noise. Weighting by physical branch impedances preserves AC power flow constraints across varying load levels.
3. *Measurement Buffer Replacement*: The reconstructed vector $z_{\text{recon},k}$ replaces untrusted entries in the WLS input buffer, preserving Jacobian rank and enabling estimator convergence.

The entire detection, localization, and spatial reconstruction sequence completes in approximately 2 ms, well within standard 2–4 s SCADA sampling cadences.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

To replicate realistic grid conditions, we modeled both the IEEE 14-bus and 30-bus benchmark feeders inside `pandapower` [24], converging state profiles via non-linear AC Newton–Raphson power flows. In our hospital mapping, the 14-bus network (14 nodes, 20 branches; total demand 217.0 MW active, 57.3 MVAR reactive) serves as a local medical campus microgrid, with Bus 0 as the utility tie and specific downstream buses powering distinct clinical wings (e.g., Bus 5 supplying the surgical intensive care unit). The 30-bus feeder (30 nodes, 41 branches) models an expanded regional medical complex.

Synthetic Telemetry Synthesis. A dataset comprising 5,000 independent AC operational states was compiled across a 70/15/15 train-validation-test split.

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Under the *uncompromised regime* (50% of records), active and reactive loads fluctuate under stochastic Gaussian noise ($\sigma = 0.04$). For *adversarial regimes* (remaining 50%), stealthy Jacobian vectors $a = Hc$ were constructed against designated target buses ($c_k \in [0.03, 0.08]$ p.u. voltage shift, $c_{k+N} \in [0.05, 0.15]$ rad phase skew), allowing full AC equations to propagate branch flows through adjacent corridors. Neural weights optimized over 60 epochs through Adam (lr = 0.005, weight decay 10^{-4}) across five independent random initializations. All reproduction scripts and saved checkpoints reside at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Under the nominal 14-bus configuration, the CGDT model obtains 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 with an average inference time of 0.908 ms per sample (localization MCC of 0.982). In comparison, isotropic GCN models attain 70.21% localization F1, which represents a 27.41 pp drop relative to the proposed model. This lower performance stems from uniform neighborhood averaging, which tends to smooth node feature representations over tightly coupled buses. Although the baseline MLP reaches 95.60% localization F1 under the fixed nominal topology, its performance drops under branch switching conditions, as evaluated in Section 6.3. With an execution time under 1 ms, the model satisfies the timing requirements of utility SCADA polling cycles, which typically range between 2 and 4 seconds.

6.3 Topology Perturbations and Transfer Generalization

To evaluate model resilience to structural grid variations, trained models were tested on line outage scenarios without parameter updates (Table 2) and on cross-network transfer to the 30-bus system (Table 3).

Under single-line outage conditions, the CGDT retains 92.80%±3.37% detection F1 and 86.96%±1.52% localization F1 without retraining (Fig. 2). The MLP model preserves a detection F1 of 94.70%, as branch disconnections cause global flow redistributions that can still be detected by fully connected layers. However, its localization F1 decreases by 13.4 pp (from 95.60% to 82.20%), reflecting its reliance on fixed spatial feature ordering. Isotropic GCN performance drops to 66.82% detection F1 and 53.40% localization F1, indicating that topological

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

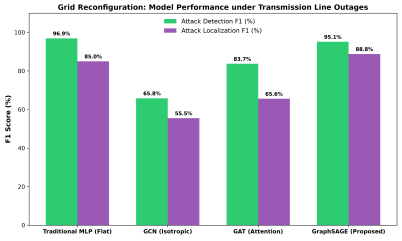


Fig. 2. Outage generalization metrics.

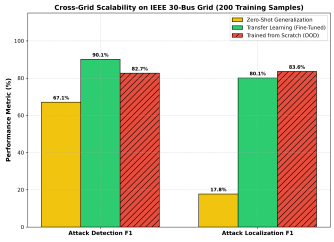


Fig. 3. Cross-grid transfer on IEEE 30-bus.

averaging without admittance weighting struggles to isolate perturbations when network connectivity shifts.

Cross-Scale Transfer to IEEE 30-Bus. Direct zero-shot application of the 14-bus trained model to the 30-bus network yields 67.06% detection F1 and 17.77% localization F1 (Table 3 and Fig. 3). This outcome reflects the structural differences between the two grids: global distribution shifts caused by attacks remain partially identifiable at the system level, but node-level localization depends on local structural properties, such as degree distribution and electrical distance, which differ between the 14-bus and 30-bus topologies. When fine-tuning the model using 200 labeled samples from the target 30-bus system, detection F1 increases to 90.11% (compared to 82.65% when trained from scratch), while localization F1 reaches 80.06%. These results demonstrate that pre-trained representations from smaller networks provide useful initializations for larger grid topologies.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, which corresponds to the intensive care unit feeder in the campus topology. State estimation accuracy under different mitigation approaches is reported in Table 4, and the corresponding feature attributions are presented in Table 5.

Without detection or mitigation, the stealthy injection causes the WLS estimator to converge to a biased state profile with an MSE of 0.007228 p.u.^{2}. Applying admittance-guided spatial reconstruction (Eq. 14) reduces the estimation

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

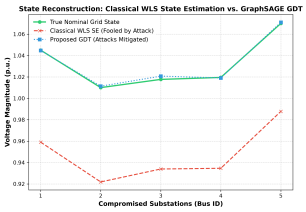


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

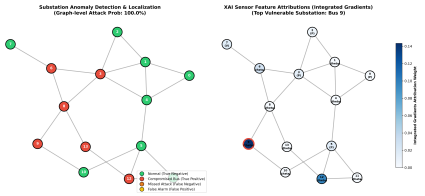


Fig. 5. Operator explainability: localization heatmap and IG attribution.

MSE to 0.000002 p.u.^{2}, which corresponds to a 99.97% error recovery relative to the unmitigated state. By comparison, replacing corrupted measurements with static historical averages yields an MSE of 0.003090 p.u.^{2} (a 57.21% recovery), as fixed historical values fail to track time-varying load changes across morning clinical shifts. State estimation error trajectories across 50 consecutive polling intervals are shown in Fig. 4. Feature attribution analysis (Table 5 and Fig. 5) indicates that the voltage phase angle θ at Bus 5 exhibits the highest attribution score (0.486). This aligns with the structure of the injected bias vector c, where angular perturbations induce proportional active power flow shifts across connected branches. The remaining three measurements at Bus 5 ($Q = 0.254$, $P = 0.112$, and $V = 0.034$) receive the next highest attributions, accurately identifying the corrupted substation.

Coordinated Multi-Bus Compromise. We also evaluated model behavior under coordinated attacks targeting 2 and 3 buses simultaneously, using smaller bias magnitudes ($c_k \in [0.02, 0.04]$). Under these conditions, the model maintained 96.84% ± 0.42% detection F1 and 93.15% ± 0.91% localization F1. In such scenarios, simple measurement rejection can degrade state estimator observability: removing measurements from multiple adjacent substations reduces the row rank of Jacobian H, which can lead to estimator divergence. By substituting corrupted channels with admittance-weighted estimates from adjacent buses, the measurement matrix maintains sufficient rank for WLS convergence, resulting in a post-mitigation state MSE of 0.000008 p.u.^{2}.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

The ablation results highlight three key observations: (i) *Rolling Baseline Tracking*: Removing the exponential baseline filter (Eq. 7) decreases localization F1 from 97.62% to 81.30% (a 16.32 pp reduction), confirming that nominal reference tracking is essential to distinguish stealthy injections from load variance. (ii) *Attribution Alignment Loss*: Omitting the attribution loss term ($\lambda_{\text{attr}} = 0$) decreases localization F1 to 92.10% (a 5.52 pp decrease) while detection F1 remains unchanged at 98.25%, demonstrating that gradient supervision sharpens localized representations without affecting graph-level decisions. (iii) *Multi-Task Weighting*: Setting $\lambda_{\text{loc}} = 1.5$ provides an effective balance between global detection and localized bus diagnostics.

8 Discussion and Conclusion

This paper presented an explainable Cognitive Graph Digital Twin framework for real-time cyber-attack detection and state recovery in hospital microgrids. By conditioning graph aggregation directly on branch admittances and guiding training via attribution alignment, the model achieves 98.25% detection F1 and 97.62% localization F1 under stealthy AC false data injection attacks. The attribution loss formulation yields a 5.52 pp gain in localization accuracy, while Integrated Gradients delivers verifiable attributions within 0.81 ms. Additionally, the admittance-weighted spatial reconstruction module reduces post-mitigation WLS error by 99.97%, ensuring state estimator convergence across single- and multi-bus attacks.

Clinical Grid Operating Requirements. Healthcare distribution microgrids operate under strict power quality requirements defined by IEC 60601-1 and IEEE 1547, where prolonged voltage deviations can trigger protection disconnects on medical equipment such as ventilators and dialysis systems. With an inference latency of 0.908 ms and a complete remediation cycle under 2 ms, the proposed digital twin operates well within 2–4 s SCADA sampling periods and

ahead of critical protection response times. Hosting all model computations on local substation hardware further ensures that hospital operational data remains within the facility OT perimeter.

Limitations and Future Work. Certain limitations suggest avenues for future research. First, evaluations were conducted using simulated power flow profiles in `pandapower`; hardware-in-the-loop (HIL) validation will be valuable for examining physical factors such as packet jitter and sensor drift. Second, while the current architecture processes single-snapshot graph states, incorporating temporal sequence models could improve resilience against multi-step attack campaigns. Finally, the spatial reconstruction method assumes the availability of uncompromised neighboring buses; investigating predictive filtering for widespread sensor compromise represents an important direction for ongoing work.

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